Spec: Primo VE ingest for full-texting of EAD finding aids

- Background
- Step 1: Retrieve new/updated XML files from s3 bucket
- Step 2: Confirm presence of Alma/Aleph number in XML, create list of EAD identifiers
  - 2A. Confirm Alma number
  - 2B. Create list of Alma numbers and corresponding EAD identifiers
- Step 3: Apply XSL
- Step 4: Compare text output to previously loaded text file
- Step 5: Copy text files to Ex Libris AWS S3 bucket
- BEGIN SECTION WITH ALMA APIS
- Step 6: Alma API to check if a Representation already exists
- Step 7: If Representation exists, delete associated file and add new file
  - 7A. Get File Details
  - 7B. Delete File
  - 7C. Post new File
- Step 8: If no Representation exists: add bib to Collection, add new Representation and new file
  - 8A. Add bib to Collection
  - 8B. Add Representation
  - 8C. Add new File
- APPENDIX: AWS command line
- FOR CORINNA ONLY: LibOW workflow: Deletes

Background

- A finding aid file is also referred to as an EAD (Encoded Archival Description). It's an XML format specific to archives. EAD files are output from the ASpace system.
- For (almost) every EAD file output by ASpace, there is a corresponding MARC record in Alma. The record in ASpace is a much longer and more detailed inventory of the collection including lists of correspondents etc., while the Alma record has some different data and fewer of the extended details.
- Since 2015 we have been indexing the full-text of the finding aids in HOLLIS to improve discovery. The finding aid does not exist as a separate record in HOLLIS – it's acts more as a full-text extension of the Alma record. You can NOT view the finding aid itself within HOLLIS, but the words from it are added to the HOLLIS search index, and a snippet is displayed in the search results.

HOLLIS search result

Step 1: Retrieve new/updated XML files from s3 bucket

- Maura set up a pipeline to publish XML files from ASpace to an s3 bucket. Credentials are in shared 1Pass
  - QA bucket: harvard-huit-lts-arckafka-pipelines-qa
  - Prod bucket: harvard-huit-lts-arckafka-pipelines
- Corinna has done the initial load of all XML files into Alma, so the script is starting with only incremental loads of new/updated files.
- I would like to implement the script with looking at only files that start with hua*, with new/changed date after Feb 2, 2026 for the initial run. This will eventually be expanded to look for all files, but I need to start with just hua* so I can monitor before adding in all the other units.
  - It might be wise to use a config file to list which 3-char units are being processed. We are starting with 'hua' and will follow with other units such as 'sch' and 'med'. Note that as units get added they won't need a special 'start' date – it will be okay to just use "check for new/updated' since last run of script
- FYI, there is also be support for web access to the files but this is primarily for staff troubleshooting

- QA: NOTE- this web page is not a reflection of what is in AWS QA bucket https://arclight-qa.lib.harvard.edu/data/eads/
- Prod: https://hollisarchives.lib.harvard.edu/data/eads/
  - This web page IS a reflection of what is in AWS prod bucket.

Step 2: Confirm presence of Alma/Aleph number in XML, create list of EAD identifiers

2A. Confirm Alma number

- We only want to process XML files that have an Aleph or Alma number in it. We want to search for any <processinfo> element containing a <head> element with any of these strings: Aleph, Alma, aleph, alma, MMS
- Example:

```
<processinfo id="aspace_ba56210917b73ea03b5b32524f039baa">
      <head>Aleph ID</head>
      <p>009588395</p>
</processinfo>
```

- There should be a numeric string in the <p> element.
- The string should be either 9 digits with a leading zero, or a number in the "Alma" format i.e. starts with 99 and ends with 3941.
- If the XML does not match the above criteria, do not do anything with it. No need to report.

2B. Create list of Alma numbers and corresponding EAD identifiers

- For each XML that meets our criteria, add the Alma/Aleph number to a list that will be included in the log of the Job Monitor.
- If the number is in the "Aleph / short form" i.e. 9-digits with leading zero, prefix it with 99 and suffix it with 0203941. For example, 009588395 will become 990095883950203941.
- To each row, add the EAD identifier to the right of the Alma/Aleph number. The EAD identifier is in the XML ead/eadheader/eadid. In the following example the value is "hua00001"

```
<ead xmlns="urn:isbn:1-931666-22-9" xmlns:xlink="http://www.w3.org/1999/xlink" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="urn:isbn:1-931666-22-9 https://www.loc.gov/ead/ead.xsd">
  <eadheader countryencoding="iso3166-1" dateencoding="iso8601" langencoding="iso639-2b" repositoryencoding="iso15511">
    <eadid countrycode="US" mainagencycode="US-MH-Ar">hua00001</eadid>
```

- The resulting log for the Job Monitor will look like this, with Alma number first then a space or comma followed by the EAD identifier:

```
990095883950203941,hua00001
990139015640203941,hua03014
```

Step 3: Apply XSL

- For the files that are new/updated and match the criteria, apply this XSL transform: Alma/discovery-ead at master · cbaksik/Alma (last update of file in GitHub should show as Feb 2, ~4:30pm)
- Output files will be simple text files, and the file extension of the output files should be changed from .xml to .txt
- It's conceivable that the XSL won't produce output for a given XML if the XML lacks certain elements, but I would not expect this to happen for any XML that had the Aleph/Alma number.

Step 4: Compare text output to previously loaded text file

- Sometimes the updated XML file will have updates that don't interest us. For this reason, we will compare the text output from the XSL with the text output from the previous version of the text. If there is no previous version of the text file, continue to Step 5.
- If there are differences in the text files, replace the file on dropbox with the new one.
- If the only differences are spaces, capitalization, punctuation, tabs, or empty lines, we will ignore the updated file and stop here for this file. Add a suffix to the log for the relevant line with "No significant differences." Move on to next file.
- Punctuation to ignore: !?()[]/:;.,"'-$%
  - i.e. exclamation, question mark, left/right parens and brackets, forward slash, colon, semicolon, period, comma, double quote single quote, hyphen, dollar, percent
- **IMPORTANT DETAIL**: the resulting text file that we load into Alma should retain the punctuation. We just want to ignore for comparison purposes. Why? Snippets will show in the Pirmo search results so we want the snippets to include the punctuation for readability, e.g.
  - Dogs are cute. Cats – not as cute unless they are kittens. "Raining cats and dogs" is not cute.
- For the initial run of the script, the previously loaded text files are on kant in home/users/cbaksik/ve-ead. Please move them to a new dir at: dropbox/alma/alma/ve-ead

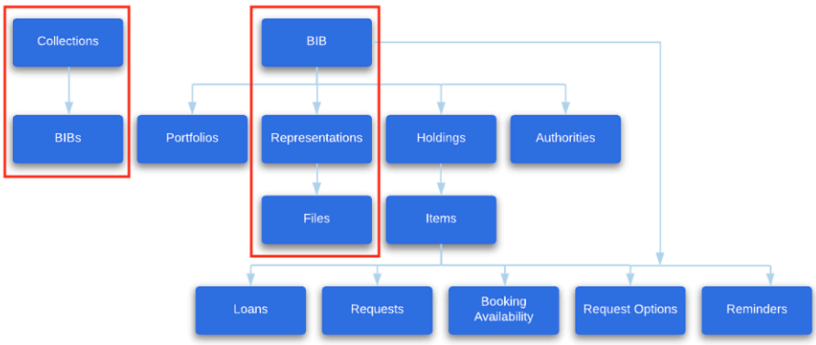
Step 5: Copy text files to Ex Libris AWS S3 bucket

- In order to use the Alma API to add files, the files first need to be in the ExL S3 bucket in this “folder”:
  - sandbox: s3://na-test-st01.ext.exlibrisgroup.com/01HVD_INST/upload/ead/
  - production: same as above except use: na-st01.ext.exlibrisgroup.com
- credentials: see 1Pass LTS-Alma “Ex Libris AWS S3 bucket (used for Alma-D EAD loads)”
- use default region and output format
- If for any reason a file of the same name exists it should be replaced
- (FYI Alma will automatically move files out of /upload and into /storage later in the process)

BEGIN SECTION WITH ALMA APIS

> The Alma API calls listed below are documented in the ‘Digital Representations’ section of the REST documentation on the ExL site: Bibliographic Records and Inventory - Ex Libris Developer Network

Brief explanation of Alma record structure relevant to these calls:



Step 6: Alma API to check if a Representation already exists

- Use the Alma API to check for an existing ‘Representation’ for the Alma identifier for each row in our log
- Sandbox example using Alma ID 99153683420303941 :
- ```
  https://api-na.hosted.exlibrisgroup.com/almaws/v1/bibs/99153683420303941/representations?apikey=
  ```
- In the response, check for a representation containing label “EAD full-text indexing.” If there is no representation, or if there is no representation containing that label, proceed to “**Step 8: Add bib to Collection, add new Representation + File.**”
- ```
  <?xml version="1.0" encoding="UTF-8" standalone="yes"?>
  <representations total_record_count="1">
      <representation is_remote="false" link="https://api-na.hosted.exlibrisgroup.com/almaws/v1/bibs/99153683420303941/representations/122703486400003941">
          <id>122703486400003941</id>
          <library desc="Online Resource">NET</library>
          <label>EAD full-text indexing</label>
          <public_note></public_note>
          <usage_type desc="Auxiliary">AUXILIARY</usage_type>
          <active desc="Active">true</active>
          <access_rights_policy_id/>
          <copyrights_statement_id/>
          <delivery_url>https://na03-psb.alma.exlibrisgroup.com/discovery/delivery/01HVD_INST:HVD2/122703486400003941</delivery_url>
          <thumbnail_url>https://na03-psb.alma.exlibrisgroup.com/view/delivery/thumbnail/01HVD_INST/122703486400003941</thumbnail_url>
          <po_line></po_line>
          <repository>Alma</repository>
          <created_by>System</created_by>
          <created_date>2025-12-04Z</created_date>
          <last_modified_by>System</last_modified_by>
          <last_modified_date>2025-12-05Z</last_modified_date>
          <files total_record_count="1" link="https://api-na.hosted.exlibrisgroup.com/almaws/v1/bibs/99153683420303941/representations/122703486400003941/files"/>
      </representation>
  </representations>
  ```
- For a sandbox example with multiple representations, where only 1 meets our criteria, see MMS ID 990075049660203941.

- If there are multiple representations that contain the label we're checking for, create a ticket and continue with the remainder of the script for the rest of the Alma numbers.
  - For a sandbox example of this scenario, see 99153775571403941

Step 7: If Representation exists, delete associated file and add new file

7A. Get File Details

For the EAD representation, use the link in the <files> element from the previous call to initiate a new call to get the file details:

```
https://api-na.hosted.exlibrisgroup.com/almaws/v1/bibs/99153683420303941/representations/122703486400003941/files?apikey=
```

There should only be 1 file per any given EAD representation.

API response for file details:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<representation_files total_record_count="1">
    <representation_file>
        <pid>132703483870003941</pid>
        <thumbnail_url>https://na03-psb.alma.exlibrisgroup.com/view/delivery/thumbnail/01HVD_INST/132703483870003941</thumbnail_url>
        <label>hua30018.txt</label>
        <size>12216</size>
        <path>01HVD_INST/storage/alma/E9/04/5F/29/C4/45/64/5D/B7/B9/BF/60/0E/E0/84/39/hua30018.txt</path>
        <fulltext>
            <fulltext_format>PLAIN</fulltext_format>
            <fulltext_url>https://na03-psb.alma.exlibrisgroup.com/view/delivery/text/01HVD_INST/132703483870003941</fulltext_url>
        </fulltext>
    </representation_file>
</representation_files>
```

7B. Delete File

Use the DELTE method to delete the file using the <pid> identifier, e.g. :

```
DELETE https://api-
na.hosted.exlibrisgroup.com/almaws/v1/bibs/99153694663703941/representations/122703486320003941/files/132703483870003941?apikey=
```

This method returns HTTP 204 (No Content) which means that the server successfully processed the request.

7C. Post new File

(Note - **Step 8C** is also Post new File so you'll be doing this one later too in the case of new EAD files.)

Now add the new file to the Representation using POST and a payload. The path provided in the payload should be to the location of the file in S3, starting with the institution code, i.e. /01HVD_INST/upload/ead/xxx99999.txt

```
POST https://api-na.hosted.exlibrisgroup.com/almaws/v1/bibs/99153694663703941/representations/122703486320003941/files?apikey=
```

Example payload:

```
{
  "label": "hua01006.txt",
  "path": "01HVD_INST/upload/ead/hua01006.txt"
}
```

Note that this API moves the file from the AWS upload folder to the storage location and therefore may take several seconds to complete, depending on the size of the file.

API response after posting new file:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<representation_file>
    <pid>132703731020003941</pid>
    <thumbnail_url>https://na03-psb.alma.exlibrisgroup.com/view/delivery/thumbnail/01HVD_INST/132703731020003941</thumbnail_url>
    <label>hua01006.txt</label>
    <size>1108</size>
    <path>01HVD_INST/storage/alma/E9/04/5F/29/C4/45/64/5D/B7/B9/BF/60/0E/E0/84/39/hua01006.txt</path>
</representation_file>
```

Step 8: If no Representation exists: add bib to Collection, add new Representation and new file

8A. Add bib to Collection

Alma will not let you add a new Representation unless the Bib is in a Collection. Add the bib to a Collection using the Collection ID (starts with 81, in the POST call below), and the Alma ID (99... 3941) in the Body.

```
POST https://api-na.hosted.exlibrisgroup.com/almaws/v1/bibs/collections/812710467680003941/bibs?apikey=
```

Body:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<bib>
  <mms_id>99155522680403941</mms_id>
  <record_format>marc21</record_format>
</bib>
```

We don't need any of the data in the success response (200).

If API returns 701115, okay to ignore and proceed to next step. Example of 701115:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<web_service_result xmlns="http://com/exlibris/urm/general/xmlbeans">
    <errorsExist>true</errorsExist>
    <errorList>
        <error>
            <errorCode>701115</errorCode>
            <errorMessage>Bib record is already assigned to this collection.</errorMessage>
            <trackingId>E01-0302205002-NPMKS-AWAE486590600</trackingId>
        </error>
    </errorList>
</web_service_result>
```

8B. Add Representation

Now that Bib is in the Collection, we can add the Representation:

```
POST https://api-na.hosted.exlibrisgroup.com/almaws/v1/bibs/99155522680403941/representations?apikey=
```

Body:

```
<representation is_remote="false" link="string">
  <library>NET</library>
  <label>EAD full-text indexing</label>
   <usage_type desc="Auxiliary">AUXILIARY</usage_type>
   <active desc="Active">true</active>
   <repository>Alma</repository>
</representation>
```

The representation ID will be returned by Alma in the response in the <id> field and this will be needed in Step 8C.

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<representation is_remote="false">
    <id>12271934417000394 1</id>
    <library desc="Online Resource">NET</library>
    <label>EAD full-text indexing</label>
    <public_note></public_note>
    <usage_type desc="Auxiliary">AUXILIARY</usage_type>
    <active desc="Active">true</active>
    <access_rights_policy_id/>
    <copyrights_statement_id/>
    <delivery_url>https://na03-psb.alma.exlibrisgroup.com/discovery/delivery/01HVD_INST:HVD2/122719344170003941</delivery_url>
    <thumbnail_url>https://na03-psb.alma.exlibrisgroup.com/view/delivery/thumbnail/01HVD_INST/122719344170003941</thumbnail_url>
    <po_line></po_line>
    <repository>Alma</repository>
    <created_by>exl_api</created_by>
    <created_date>2026-03-02Z</created_date>
    <last_modified_by>API, corinna QA troubleshooting key</last_modified_by>
    <last_modified_date>2026-03-02Z</last_modified_date>
    <files total_record_count="0" link="https://api-na.hosted.exlibrisgroup.com/almaws/v1/bibs/99153725220203941/representations/122719344170003941/files"/>
</representation>
```

8C. Add new File

This is the same as Step 7C.

APPENDIX: AWS command line

Corinna's notes

aws configure
AWS Access Key ID [**********TGGZ]:
AWS Secret Access Key [**********HywV]:
Default region name [us-east-1]:
Default output format [text]:

[to upload]

aws s3 cp hua01006.txt s3://na-test-st01.ext.exlibrisgroup.com/01HVD_INST/upload/ead/

FOR CORINNA ONLY: LibOW workflow: Deletes

from maura: what were going to do is have a deletes folder where we 'move' the deleted items, and it will just be like orc00001.xml and contains
<eadid>orc00001</eadid>and this folder will only retain items for 30 days

this way you have files with timestamps for processing but the data is actually purged, and 30 days seems reasonable